

Chapter 9: Inferences Based on Two Samples

Probability and Statistics for Science and Engineering
with Examples in R

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(1) Inferences Concerning Two means

Setting:

1. $X_1, \dots, X_m$ is a random sample of size m from population 1 which has mean μ_1 and variance σ_1^2 .
2. $Y_1, \dots, Y_n$ is a random sample of size n from population 2 which has mean μ_2 and variance σ_2^2 .
3. The two samples $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ are independent.

Inferences will be made about the difference in means:

$$\mu_1 - \mu_2$$

(A) Large, Independent Samples

(Both $m, n \geq 30$)

$$\begin{aligned} 1. E(\bar{X} - \bar{Y}) &= E(\bar{X}) - E(\bar{Y}) = \mu_1 - \mu_2 = \Delta \\ 2. Var(\bar{X} - \bar{Y}) &= Var(\bar{X}) + Var(\bar{Y}) = \frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n} \end{aligned} \quad \left. \vphantom{\begin{aligned} 1. E(\bar{X} - \bar{Y}) &= E(\bar{X}) - E(\bar{Y}) = \mu_1 - \mu_2 = \Delta \\ 2. Var(\bar{X} - \bar{Y}) &= Var(\bar{X}) + Var(\bar{Y}) = \frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n} \end{aligned}} \right\} \begin{array}{l} \text{CLT} \\ \uparrow \\ \text{Independence} \end{array} Z = \frac{(\bar{X} - \bar{Y}) - \Delta}{\sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}}} \rightarrow N(0,1)$$

statistic and its distribution:

When the sample sizes m and n are large ($m, n \geq 30$), the normal approximation remains valid when sample variances replace the population variances. Consequently, the two sample Z statistic becomes

$$Z = \frac{(\bar{X} - \bar{Y}) - \Delta}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}} \rightarrow N(0,1).$$

(a) CI

$$P \left(-z_{\alpha/2} < \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}}} < z_{\alpha/2} \right) \approx 1 - \alpha$$

$$P \left(-z_{\alpha/2} \sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}} < (\bar{X} - \bar{Y}) - (\mu_1 - \mu_2) < z_{\alpha/2} \sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}} \right) \approx 1 - \alpha$$

$$P \left((\bar{X} - \bar{Y}) - z_{\alpha/2} \sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}} < \mu_1 - \mu_2 < (\bar{X} - \bar{Y}) + z_{\alpha/2} \sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}} \right) \approx 1 - \alpha$$

For two independent populations, a $100(1 - \alpha)\%$ CI for $\mu_1 - \mu_2$:

$$\left((\bar{x} - \bar{y}) - z_{\frac{\alpha}{2}} \sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}}, (\bar{x} - \bar{y}) + z_{\alpha/2} \sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}} \right)$$

If $m \geq 30$ and $n \geq 30$.

Example 9.1: In June 2023, chemical analyses were made of 85 water samples (each of unit volume) taken from various parts of a city lake, and the measurements of chlorine content(含氯量) were recorded. During the next two winters, the use of road salt was substantially reduced in the catchment areas(集水区域) of the lake. In June 2025, 110 water samples were analyzed.

	Chlorine Content (in ppm)	
	2023	2025
Mean	18.3	17.8
Standard Deviation	1.2	1.8

Construct a 95% CI for the difference of the population means.

Since both $m = 85$ and $n = 110$ are large, we can use the following formula.

$$\begin{aligned}
 \bar{x} - \bar{y} \pm z_{\alpha/2} \sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}} &= 18.3 - 17.8 \pm z_{0.025} \sqrt{\frac{(1.2)^2}{85} + \frac{(1.8)^2}{110}} \\
 &= 0.5 \pm 1.96(0.2154) = 0.5 \pm 0.422
 \end{aligned}$$

Thus, a 95% confidence interval for $\mu_1 - \mu_2$ is (0.078, 0.922).

0 is not included $\rightarrow$ significant difference

Testing hypothesis

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: \mu_1 - \mu_2 = \Delta_0,$ $H_1: \mu_1 - \mu_2 \neq \Delta_0$ (2-sided alternative)	$H_0: \mu_1 - \mu_2 = \Delta_0,$ $H_1: \mu_1 - \mu_2 > \Delta_0$ (1-sided alternative)	$H_0: \mu_1 - \mu_2 = \Delta_0,$ $H_1: \mu_1 - \mu_2 < \Delta_0$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$z = \frac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$ <u>Rejection Region:</u> $ z \geq z_{\alpha/2}$	$z = \frac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$ <u>Rejection Region:</u> $z \geq z_{\alpha}$	$z = \frac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$ <u>Rejection Region:</u> $z \leq -z_{\alpha}$
Step 4	Substitute $\bar{x}, \bar{y},$ s_1^2, s_2^2, m and n Calculate z Decision	Substitute $\bar{x}, \bar{y}, s_1^2, s_2^2, m$ and n Calculate z Decision	Substitute $\bar{x}, \bar{y},$ s_1^2, s_2^2, m and n Calculate z Decision
Step 5	$p = 2P(Z \geq z)$	$p = P(Z \geq z)$	$p = P(Z \leq z)$

Example 9.2: From Example 9.1, do the data provide strong evidence that **there is a reduction of average chlorine level** in the lake water in 2025 compared to the level in 2023? Test with $\alpha = 0.05$.

$$H_0: \mu_1 = \mu_2, \text{ } H_1: \mu_1 > \mu_2$$
$$\alpha = 0.05$$

$$\text{Test statistic: } Z = \frac{\bar{X} - \bar{Y} - 0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}} \rightarrow N(0,1)$$

$$\text{Rejection region: } z \geq z_\alpha = z_{0.05} = 1.645$$

$$z = \frac{18.3 - 17.8}{\sqrt{\frac{(1.2)^2}{85} + \frac{(1.8)^2}{110}}} = 2.32 > 1.645$$

Reject H_0

→ Yes, the data provide strong evidence that there is a reduction of average chlorine level in the lake water in 2025 compared to 2023.

$$p\text{-value} = P(Z > 2.32) = 1 - \Phi(2.32) = 0.01$$

Using R to find P-value, `>1-pnorm(2.32)`

(B) Small, Independent Samples

($m < 30$ and/or $n < 30$)

Assumption: Both populations are normal

1. $X_1, \dots, X_m$ is a random sample of size m from normal population 1 which has mean μ_1 and variance σ_1^2 .
2. $Y_1, \dots, Y_n$ is a random sample of size n from normal population 2 which has mean μ_2 and variance σ_2^2 .
3. The two samples $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ are independent.

Inferences will be made about the difference in means:

$$\mu_1 - \mu_2$$

(a) When we assume $\sigma_1 = \sigma_2 = \sigma$

$$\left. \begin{array}{l} 1. E(\bar{X} - \bar{Y}) = E(\bar{X}) - E(\bar{Y}) = \mu_1 - \mu_2 = \Delta \\ 2. \text{Var}(\bar{X} - \bar{Y}) \overset{\text{Independence}}{=} \text{Var}(\bar{X}) + \text{Var}(\bar{Y}) = \sigma^2 \left(\frac{1}{m} + \frac{1}{n} \right) \end{array} \right\} Z = \frac{(\bar{X} - \bar{Y}) - \Delta}{\sigma \sqrt{\left(\frac{1}{m} + \frac{1}{n} \right)}} \sim N(0,1)$$

The unknown σ^2 is estimated by the following pooled estimator:

$$S_p^2 = \frac{(m-1)S_1^2 + (n-1)S_2^2}{m+n-2}$$

Test statistic:

$$T = \frac{(\bar{X} - \bar{Y}) - \Delta}{S_p \sqrt{\left(\frac{1}{m} + \frac{1}{n} \right)}} \sim t_{m+n-2}$$

where $S_p^2 = \frac{(m-1)S_1^2 + (n-1)S_2^2}{m+n-2}$

For two independent populations, a $100(1 - \alpha)\%$ CI for $\mu_1 - \mu_2$:

$$\left(\bar{x} - \bar{y} - t_{\frac{\alpha}{2}, m+n-2} s_p \sqrt{\frac{1}{m} + \frac{1}{n}}, \bar{x} - \bar{y} + t_{\frac{\alpha}{2}, m+n-2} s_p \sqrt{\frac{1}{m} + \frac{1}{n}} \right)$$

If $m < 30$ and/or $n < 30$

Example 9.3: A random sample of 15 mothers with low socioeconomic status deliver babies whose average birthweight was 110 ounces and a sample standard deviation was 24 ounces, whereas a random sample of 76 mothers with a medium socioeconomic status resulted in a sample average birthweight and sample standard deviation of 120 ounces and 22 ounces, respectively. Assume that the average birthweights are normally distributed in both populations. Find a 95% CI of $\mu_1 - \mu_2$.

$$s_p^2 = \frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2} = \frac{14(24)^2 + 75(22)^2}{15+76-2} = 498.5$$

$$t_{\frac{\alpha}{2}, m+n-2} = t_{0.025, 89} = 1.99$$

$$\bar{x} - \bar{y} \pm t_{\frac{\alpha}{2}, m+n-2} s_p \sqrt{\frac{1}{m} + \frac{1}{n}} = 110 - 120 \pm 1.99 \sqrt{498.5} \sqrt{\frac{1}{15} + \frac{1}{76}} = -10 \pm 12.5$$

95% confidence interval for $\mu_1 - \mu_2$: $(-22.5, 2.5)$

Testing hypothesis

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: \mu_1 - \mu_2 = \Delta_0,$ $H_1: \mu_1 - \mu_2 \neq \Delta_0$ (2-sided alternative)	$H_0: \mu_1 - \mu_2 = \Delta_0,$ $H_1: \mu_1 - \mu_2 > \Delta_0$ (1-sided alternative)	$H_0: \mu_1 - \mu_2 = \Delta_0,$ $H_1: \mu_1 - \mu_2 < \Delta_0$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$t = \frac{(\bar{x} - \bar{y}) - \Delta_0}{s_p \sqrt{\frac{1}{m} + \frac{1}{n}}}$ Rejection Region: $t \geq t_{\frac{\alpha}{2}, m+n-2}$	$t = \frac{(\bar{x} - \bar{y}) - \Delta_0}{s_p \sqrt{\frac{1}{m} + \frac{1}{n}}}$ Rejection Region: $t \geq t_{\alpha, m+n-2}$	$t = \frac{(\bar{x} - \bar{y}) - \Delta_0}{s_p \sqrt{\frac{1}{m} + \frac{1}{n}}}$ Rejection Region: $t \leq -t_{\alpha, m+n-2}$
Step 4	Substitute $\bar{x}, \bar{y}, s_p, m$ and n Calculate t Decision	Substitute $\bar{x}, \bar{y}, s_p, m$ and n Calculate t Decision	Substitute $\bar{x}, \bar{y},$ s_p, m and n Calculate t Decision
Step 5	$p = 2P(T \geq t)$	$p = P(T \geq t)$	$p = P(T \leq t)$

Example 9.4: IQ's of 16 students from one area of a city: $\bar{x} = 107$, $s_1 = 10$; IQ's of 14 students from another area of the city: $\bar{y} = 112$, $s_2 = 8$. The IQ scores are assumed to be normally distributed. Is there a significant difference between the IQ's of the two groups at significance level of 0.01?

$$1) H_0: \mu_1 = \mu_2, H_1: \mu_1 \neq \mu_2$$

$$2) \alpha = 0.01$$

$$s_p^2 = \frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2} = \frac{15(10)^2 + 13(8)^2}{16+14-2} = 83.3$$

$$3) \text{ Test statistic: } T = \frac{\bar{X} - \bar{Y}}{s_p \sqrt{\frac{1}{m} + \frac{1}{n}}} \sim t(n+m-2)$$

$$\text{Rejection region: } |t| \geq t_{\frac{\alpha}{2}, m+n-2} = t_{0.005, 28} = 2.763$$

$$4) t = \frac{107-112-0}{\sqrt{83.3} \sqrt{\frac{1}{16} + \frac{1}{14}}} = -1.496$$

Since $|t| = 1.496 < 2.763$, we do not reject H_0 . There is not a significant difference between the IQ's of the two groups at level 0.01.

$$5) t_{0.1, 28} = 1.313 < 1.496 < 1.701 = t_{0.05, 28}$$

$$0.1 = 2 \times 0.05 < p\text{-value} < 2 \times 0.1 = 0.2$$

Example 9.4 (continued)

To calculate the p-value using R,

```
>2*(1-pt(1.496, 28))
```

```
0.1458411
```

To conduct hypothesis testing using R,

```
>t.test(x, y, conf.level=0.01,var.equal=T)
```

The output contains a 99% confidence interval for the mean difference, the value of the t statistic and the *p*-value.

(b) When we assume $\sigma_1 \neq \sigma_2$

$$\text{CI: } \bar{x} - \bar{y} \pm t_{\frac{\alpha}{2}, \nu} \sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}$$

$$\text{where } \nu = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}} \text{ (round down)}$$

$$\text{Test statistic: } T = \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}} \sim t \text{ with df } \nu = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}}$$

For two independent normal populations with $\sigma_1 \neq \sigma_2$, a $100(1 - \alpha)\%$ CI for $\mu_1 - \mu_2$:

$$\left(\bar{x} - \bar{y} - t_{\frac{\alpha}{2}, v} \sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}, \bar{x} - \bar{y} + t_{\frac{\alpha}{2}, v} \sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}} \right), \text{ where } v = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}}$$

Example 9.5: Breaking load (pound/inch width) for two kinds of fabrics
两种织物在每英寸宽度上的断裂负荷（单位为磅）

Fabric type	Sample Size	Sample Mean	Sample sd
Cotton	10	25.9	0.4
Triacetate 三醋酸纤维	10	68.1	1.8

Assuming normality, find a 95% CI for the difference between true average porosity (多孔性) for the two types of fabric.

$$v = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}} = \frac{\left(\frac{(0.4)^2}{10} + \frac{(1.8)^2}{10}\right)^2}{\frac{((0.4)^2/10)^2}{9} + \frac{((1.8)^2/10)^2}{9}} = 9.89 \rightarrow v = 9$$

$$\bar{x} - \bar{y} \pm t_{0.025, 9} \sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}} = 25.9 - 68.1 \pm 2.262 \sqrt{\frac{(0.4)^2}{10} + \frac{(1.8)^2}{10}} = -42.2 \pm 1.32$$

95% CI: $(-43.52, -40.88)$

Testing hypothesis($n < 30$)

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: \mu_1 - \mu_2 = \Delta_0$, $H_1: \mu_1 - \mu_2 \neq \Delta_0$ (2-sided alternative)	$H_0: \mu_1 - \mu_2 = \Delta_0$, $H_1: \mu_1 - \mu_2 > \Delta_0$ (1-sided alternative)	$H_0: \mu_1 - \mu_2 = \Delta_0$, $H_1: \mu_1 - \mu_2 < \Delta_0$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$t = \frac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$ Rejection Region: $ t \geq t_{\frac{\alpha}{2}, v}$ $v = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}}$	$t = \frac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$ Rejection Region: $t \geq t_{\alpha, v}$ $v = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}}$	$t = \frac{(\bar{x} - \bar{y}) - \Delta_0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$ Rejection Region: $t \leq -t_{\alpha, v}$ $v = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}}$
Step 4	Substitute $\bar{x}$, $\bar{y}$, s_1^2 , s_2^2 , m and n Calculate t Decision	Substitute $\bar{x}$, $\bar{y}$, s_1^2 , s_2^2 , m and n Calculate t Decision	Substitute $\bar{x}$, $\bar{y}$, s_1^2 , s_2^2 , m and n Calculate t Decision
Step 5	$p = 2P(T \geq t)$	$p = P(T \geq t)$	$p = P(T \leq t)$

Example 9.6: The lengths of possession (in months) by first owner were observed for mid-size sedan models from two auto companies. The company making Model B cars claims that the owners like its cars **better than the cars** of Model A.

Model A	55	54	53	56	50	63	65	62	64	55
Model B	60	67	67	65	62	58	58	58		

Assume **normality** and $\sigma_1 \neq \sigma_2$, carry out a test to see whether the data support this conclusion using $\alpha = 0.05$.

Solution: $m = 10$, $n = 8$: small sample,

$$\bar{x} = 57.70, \bar{y} = 61.88, s_1 = 5.29, s_2 = 3.98$$

$$H_0: \mu_1 - \mu_2 = 0, H_1: \mu_1 - \mu_2 < 0$$

Example 9.6 (continued)

$m = 10, n = 8$: small, $\bar{x} = 57.70, \bar{y} = 61.88, s_1 = 5.29, s_2 = 3.98$

$H_0: \mu_1 - \mu_2 = 0, H_1: \mu_1 - \mu_2 < 0$

$\alpha = 0.05$

$$v = \frac{\left(\frac{s_1^2}{m} + \frac{s_2^2}{n}\right)^2}{\frac{(s_1^2/m)^2}{m-1} + \frac{(s_2^2/n)^2}{n-1}} = \frac{\left(\frac{(5.29)^2}{10} + \frac{(3.98)^2}{8}\right)^2}{\frac{((5.29)^2/10)^2}{9} + \frac{((3.98)^2/8)^2}{7}} = 15.97 \rightarrow v = 15$$

Test statistic: $T = \frac{\bar{X} - \bar{Y} - 0}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}} \sim t(v)$

Rejection region: $t < -t_{\alpha, v} = -t_{0.05, 15} = -1.753$

$$t = \frac{57.70 - 61.88}{\sqrt{\frac{(5.29)^2}{10} + \frac{(3.98)^2}{8}}} = -1.912 < -1.753 \rightarrow \text{Reject } H_0$$

→ There is a significant difference between the lengths of time that owners keep cars of Model B compared to Model A at level 0.05 (Model B cars are kept for longer) .

$$t_{0.05, 15} = 1.753 < 1.912 < 2.131 = t_{0.025, 15}$$

$$0.025 < p\text{-value} < 0.05$$

Example 9.6 (continued)

To calculate the p -value using R,

```
>pt(-1.912, 15)
```

```
0.03758497
```

To conduct hypothesis testing using R,

```
>x=c(55, 54, 53, 56, 50, 63, 65, 62, 64, 55)
```

```
>y=c(60, 67, 67, 65, 62, 58, 58, 58)
```

```
>t.test(x, y, alt="less", var.equal=F)
```

To construct a 95% CI in Example 9.5 using R,

```
>t.test(x, y, var.equal=F)
```

(C) Paired Data

For paired random variable (X_i, Y_i) , $i=1, \dots, n$, let

$$d_i = X_i - Y_i, \quad \mu_d = \delta = E(d_i)$$

$\bar{d}, S_d$ is the sample mean and sample standard variance of $d_i = X_i - Y_i$ which is the difference of observed paired data (X_i, Y_i) , $i=1, \dots, n$.

the statistic and its distribution of CI is : $\mu_d = \delta = E(d_i)$

$$t = \frac{\bar{d} - \mu_d}{s_d / \sqrt{n}} \sim t(n-1)$$

For paired data, a $100(1 - \alpha)\%$ confidence interval for the mean difference δ :

$$\left(\bar{d} - t_{\frac{\alpha}{2}, n-1} \frac{s_d}{\sqrt{n}}, \quad \bar{d} + t_{\frac{\alpha}{2}, n-1} \frac{s_d}{\sqrt{n}} \right) \quad \text{for } n < 30$$

(C) Paired Data

For paired data, a $100(1 - \alpha)\%$ confidence interval for the mean difference δ :

$$\left(\bar{d} - t_{\alpha/2, n-1} \frac{s_d}{\sqrt{n}}, \bar{d} + t_{\alpha/2, n-1} \frac{s_d}{\sqrt{n}} \right) \text{ for } n < 30$$

Example 9.7: Two different methods for determining chlorine content(含氯量) (mg/L) on samples of Cl_2 -demand-free water for various doses & contact times.. 使用"干净"的水样，通过改变加氯量和反应时间，比较两种测氯方法的准确性

	Subject							
	1	2	3	4	5	6	7	8
Method I	0.39	0.84	1.76	3.35	4.69	7.70	10.52	10.92
Method II	0.36	1.35	2.56	3.92	5.35	8.33	10.70	10.91
d	0.03	-0.51	-0.80	-0.57	-0.66	-0.63	-0.18	0.01

$$\bar{d} = -0.414, s_d = 0.321$$

(a) A 99% CI for the difference δ in true average chlorine readings between 2 methods:

$$\bar{d} \pm t_{\alpha/2, n-1} \frac{s_d}{\sqrt{n}} = \bar{d} \pm t_{0.005, 7} \frac{s_d}{\sqrt{n}} = -0.414 \pm 3.499 \frac{0.321}{\sqrt{8}} = -0.414 \pm 0.397$$

99% CI: $(-0.811, -0.017)$

(b) Is there a significant difference in the true average chlorine readings?

0 is not included in the CI $\rightarrow$ There is a significant difference in true average.

Hypothesis testing

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: \delta = 0, H_1: \delta \neq 0$ (2-sided alternative)	$H_0: \delta = 0, H_1: \delta > 0$ (1-sided alternative)	$H_0: \delta = 0, H_1: \delta < 0$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$

If $n < 30$, conduct the t -test with $n - 1$ degrees of freedom

Step 3	$t = \frac{\bar{d}}{s_d/\sqrt{n}} \sim t(n-1)$ <u>Rejection Region:</u> $ t \geq t_{\alpha/2, n-1}$	$t = \frac{\bar{d}}{s_d/\sqrt{n}} \sim t(n-1)$ <u>Rejection Region:</u> $t \geq t_{\alpha, n-1}$	$t = \frac{\bar{d}}{s_d/\sqrt{n}} \sim t(n-1)$ <u>Rejection Region:</u> $t \leq -t_{\alpha, n-1}$
Step 4	Substitute $\bar{d}$, s_d and n Calculate t Decision	Substitute $\bar{d}$, s_d and n Calculate t Decision	Substitute $\bar{d}$, s_d and n Calculate t Decision
Step 5	$p = 2P(T \geq t)$	$p = P(T \geq t)$	$p = P(T \leq t)$

Example 9.8: A test is designed to compare the wearing qualities 耐磨性 of two brands of motorcycle tires. One tire of each brand is placed on each of 6 motorcycles that are driven for a specified mileage. The two tires are then exchanged (front and rear) and driven for an equal number of miles. Wear is measured (in thousandths of an inch). Test for difference of wearing using $\alpha = 0.05$.

	Motorcycle					
	1	2	3	4	5	6
Brand A	98	61	38	117	88	109
Brand B	102	60	46	125	94	111
d	-4	1	-8	-8	-6	-2

$$n = 6, \bar{d} = -4.5, s_d = 3.56$$

$$H_0: \delta = 0, H_1: \delta \neq 0$$

$$\alpha = 0.05$$

$$\text{Test statistic: } T = \frac{\bar{d}}{s_d/\sqrt{n}}$$

$$\text{Rejection region: } |t| \geq t_{\frac{\alpha}{2}, n-1} = t_{0.025, 5} = 2.571$$

$$t = \frac{-4.5}{3.56/\sqrt{6}} = -3.096$$

Since $|t| = 3.096 > 2.571$, we reject $H_0 \rightarrow$ There is a significant difference at level 0.05.

$$t_{0.025, 5} = 2.571 < 3.096 < 3.365 = t_{0.01, 5}$$

$$0.01 \times 2 = 0.02 < p\text{-value} < 0.05 = 0.025 \times 2$$

Example 9.8 (continued)

To calculate the p-value using R,

```
>2*pt(-3.096, 5)
```

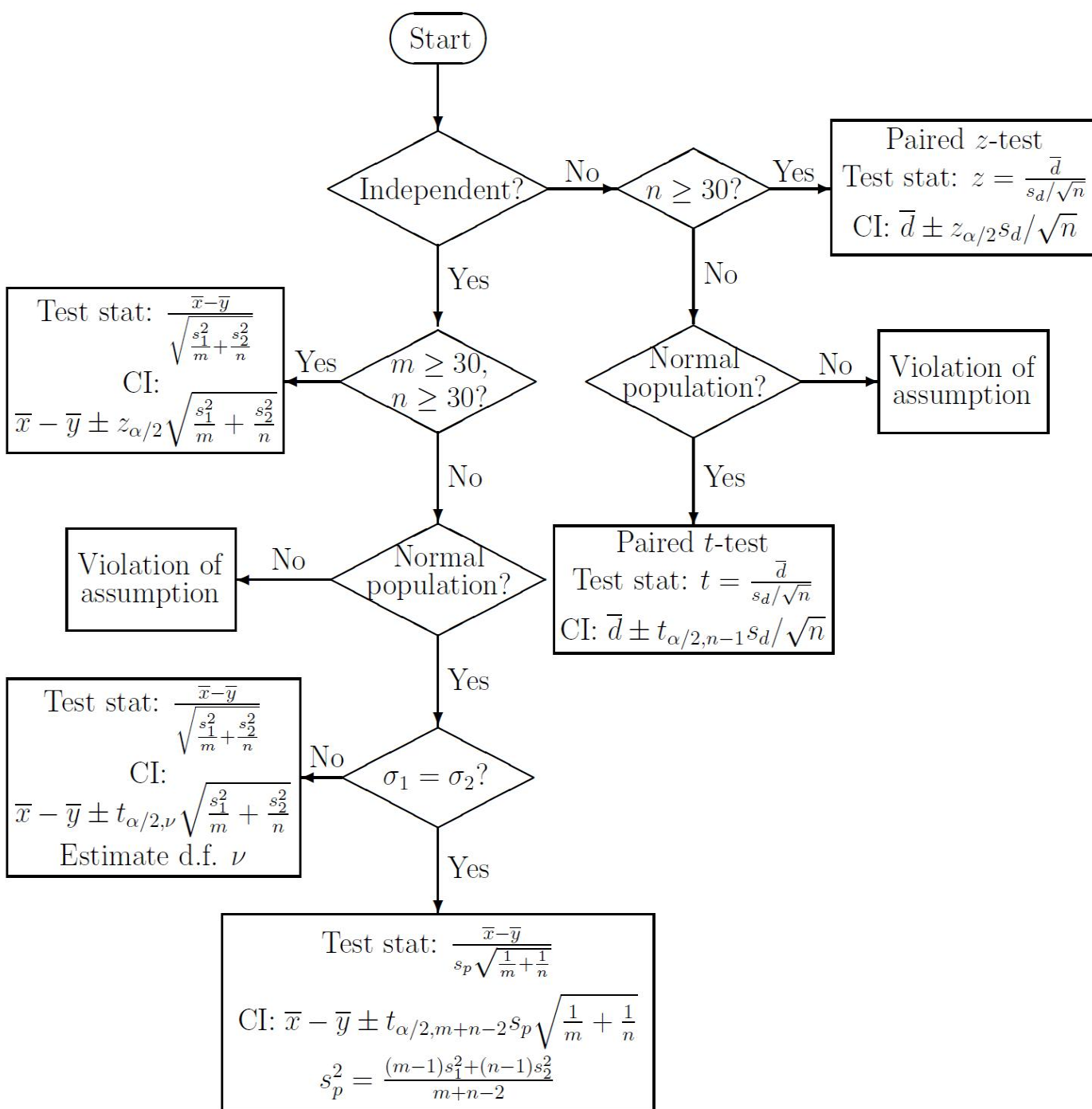
```
0.02697522
```

To conduct hypothesis testing using R,

```
>t.test(x-y)
```

Alternatively,

```
>t.test(x, y, paired=T)
```



(2) Inferences Concerning Two Population Proportions

$$X \sim \text{Bin}(m, p_1), Y \sim \text{Bin}(n, p_2)$$

$$\hat{p}_1 = x/m, \hat{p}_2 = y/n \quad E(\hat{p}_1) = E\left(\frac{X}{m}\right) = p_1, E(\hat{p}_2) = E\left(\frac{Y}{n}\right) = p_2$$

$$E(\hat{p}_1 - \hat{p}_2) = E\left(\frac{X}{m} - \frac{Y}{n}\right) = E\left(\frac{X}{m}\right) - E\left(\frac{Y}{n}\right) = p_1 - p_2$$

$$\text{Var}\left(\frac{X}{m} - \frac{Y}{n}\right) = \text{Var}\left(\frac{X}{m}\right) + \text{Var}\left(\frac{Y}{n}\right) = \frac{p_1(1-p_1)}{m} + \frac{p_2(1-p_2)}{n}$$

By CLT,

$$Z = \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{m} + \frac{\hat{p}_2(1-\hat{p}_2)}{n}}} \rightarrow \text{N}(0,1)$$

is approximately standard normal.

CI for $p_1 - p_2$

$$P\left(-z_{\alpha/2} < \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{m} + \frac{\hat{p}_2(1-\hat{p}_2)}{n}}} < z_{\alpha/2}\right) \approx 1 - \alpha$$

$$P\left(-z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{m} + \frac{\hat{p}_2(1-\hat{p}_2)}{n}} < \hat{p}_1 - \hat{p}_2 - (p_1 - p_2) < z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{m} + \frac{\hat{p}_2(1-\hat{p}_2)}{n}}\right) \approx 1 - \alpha$$

$$P\left(\hat{p}_1 - \hat{p}_2 - z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{m} + \frac{\hat{p}_2(1-\hat{p}_2)}{n}} < p_1 - p_2 < \hat{p}_1 - \hat{p}_2 + z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{m} + \frac{\hat{p}_2(1-\hat{p}_2)}{n}}\right) \approx 1 - \alpha$$

A $100(1 - \alpha)\%$ confidence interval for $p_1 - p_2$ is

$$\left(\hat{p}_1 - \hat{p}_2 - z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{m} + \frac{\hat{p}_2(1-\hat{p}_2)}{n}}, \hat{p}_1 - \hat{p}_2 + z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{m} + \frac{\hat{p}_2(1-\hat{p}_2)}{n}}\right),$$

where $\hat{p}_1 = x/m$ and $\hat{p}_2 = y/n$.

Example 9.9: A summary of data on incidence of lung cancers among smokers and nonsmokers in a certain area

	Smoker	Nonsmoker
Sample size	1548	3176
Number of lung cancers	42	24

Find a 99% CI for the difference between the two proportions.

$$m = 1548, x = 42, n = 3176, y = 24$$

$$\hat{p}_1 = \frac{x}{m} = \frac{42}{1548} = 0.0271, \hat{p}_2 = \frac{y}{n} = \frac{24}{3176} = 0.0076$$

$$\begin{aligned} & \hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{m} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n}} \\ &= 0.0271 - 0.0076 \pm z_{0.005} \sqrt{\frac{(0.0271)(0.9729)}{1548} + \frac{(0.0076)(0.9924)}{3176}} \\ &= 0.0195 \pm 2.58 \cdot 0.0044 = 0.0195 \pm 0.0114 \\ & \text{99\% CI for } p_1 - p_2: (0.0081, 0.0309) \end{aligned}$$

Hypothesis testing

Assume $p_1 = p_2 = p$ under H_0

The common proportion is estimated by

$$\hat{p} = \frac{x + y}{m + n} = \frac{m}{m + n} \hat{p}_1 + \frac{n}{m + n} \hat{p}_2$$

Hypothesis testing

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: p_1 - p_2 = 0,$ $H_1: p_1 - p_2 \neq 0$ (2-sided alternative)	$H_0: p_1 - p_2 = 0,$ $H_1: p_1 - p_2 > 0$ (1-sided alternative)	$H_0: p_1 - p_2 = 0,$ $H_1: p_1 - p_2 < 0$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{m} + \frac{1}{n} \right)}}$ <u>Rejection Region:</u> $ z \geq z_{\frac{\alpha}{2}}$	$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{m} + \frac{1}{n} \right)}}$ <u>Rejection Region:</u> $z \geq z_{\alpha}$	$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{m} + \frac{1}{n} \right)}}$ <u>Rejection Region:</u> $z \leq -z_{\alpha}$
Step 4	Substitute $\hat{p}_1, \hat{p}_2, \hat{p}, m$ and n Calculate z Decision	Substitute $\hat{p}_1, \hat{p}_2, \hat{p}, m$ and n Calculate z Decision	Substitute $\hat{p}_1, \hat{p}_2, \hat{p}, m$ and n Calculate z Decision
Step 5	$p = 2P(Z \geq z)$	$p = P(Z \geq z)$	$p = P(Z \leq z)$

Example 9.10: A certain washing machine manufacturer claims that the **fraction p_1** of his washing machines that need repairs in the first five years of operation **is less than the fraction p_2** of another brand. From 200 machines of each brand, 21 machines of his brand and 37 machines of the other brand need repairs. Do these data support the manufacturer's claim? Use $\alpha = 0.05$.

$$m = n = 200, x = 21, y = 37, \hat{p}_1 = \frac{x}{m} = \frac{21}{200} = 0.105, \hat{p}_2 = \frac{y}{n} = \frac{37}{200} = 0.185$$

$$\hat{p} = \frac{x + y}{m + n} = \frac{21 + 37}{200 + 200} = 0.145$$

$$H_0: p_1 = p_2, H_1: p_1 < p_2$$

$$\alpha = 0.05$$

$$\text{Test statistic: } Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{m} + \frac{1}{n} \right)}} \sim N(0, 1)$$

$$\text{Rejection region: } z \leq -z_\alpha = -z_{0.05} = -1.645$$

$$z = \frac{0.105 - 0.185}{\sqrt{(0.145)(0.855) \left(\frac{1}{200} + \frac{1}{200} \right)}} = -2.27 < -1.645 \rightarrow \text{Reject } H_0$$

$$p\text{-value} = P(Z < -2.27) = 0.012$$

```
In R, > a=rbind(c(21,179),c(37,163))    # 200-21=179; 200-37=163
> chisq.test(a)
```

(3) Inferences Concerning Two Variances

Independent random samples of size m and n

- **Normal populations**, assume equal variance

$$F = \frac{S_1^2}{S_2^2} \sim F_{m-1, n-1}$$

$$F_{1-\alpha, v_1, v_2} = \frac{1}{F_{\alpha, v_2, v_1}}$$

$X_1, X_2, \dots, X_m$: random sample from $N(\mu_1, \sigma_1^2)$

$Y_1, Y_2, \dots, Y_n$: random sample from $N(\mu_2, \sigma_2^2)$

The X and Y samples **are independent from one another**, then

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{m-1, n-1}$$

CI for σ_1^2 / σ_2^2

$$P\left(F_{1-\frac{\alpha}{2}, m-1, n-1} < \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} < F_{\frac{\alpha}{2}, m-1, n-1}\right) = 1 - \alpha$$

$$\left(\frac{S_1^2/S_2^2}{F_{\frac{\alpha}{2}, m-1, n-1}} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2/S_2^2}{F_{1-\frac{\alpha}{2}, m-1, n-1}}\right) = 1 - \alpha$$

$$P\left(\frac{S_1^2/S_2^2}{F_{\frac{\alpha}{2}, m-1, n-1}} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2}{S_2^2} F_{\frac{\alpha}{2}, n-1, m-1}\right) = 1 - \alpha$$

A $100(1 - \alpha)\%$ confidence interval for σ_1^2 / σ_2^2 is

$$\left(\frac{s_1^2/s_2^2}{F_{\frac{\alpha}{2}, m-1, n-1}}, \quad \frac{s_1^2}{s_2^2} F_{\frac{\alpha}{2}, n-1, m-1}\right)$$

Hypothesis testing

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: \sigma_1^2 = \sigma_2^2,$ $H_1: \sigma_1^2 \neq \sigma_2^2$ (2-sided alternative)	$H_0: \sigma_1^2 = \sigma_2^2,$ $H_1: \sigma_1^2 > \sigma_2^2$ (1-sided alternative)	$H_0: \sigma_1^2 = \sigma_2^2,$ $H_1: \sigma_1^2 < \sigma_2^2$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$f = \frac{s_1^2}{s_2^2} \sim F(m-1, n-1)$ <u>Rejection Region:</u> $f \geq F_{\frac{\alpha}{2}, m-1, n-1}$ or $f \leq 1/F_{\frac{\alpha}{2}, n-1, m-1}$	$f = \frac{s_1^2}{s_2^2} \sim F(m-1, n-1)$ <u>Rejection Region:</u> $f \geq F_{\alpha, m-1, n-1}$	$f = \frac{s_1^2}{s_2^2} \sim F(m-1, n-1)$ <u>Rejection Region:</u> $f \leq 1/F_{\alpha, n-1, m-1}$
Step 4	Substitute s_1^2 and s_2^2 Calculate f Decision	Substitute s_1^2 and s_2^2 Calculate f Decision	Substitute s_1^2 and s_2^2 Calculate f Decision
Step 5	$p = 2\min\{P(F \geq f),$ $P(F \leq f)\}$ where $F \sim F_{m-1, n-1}$	$p = P(F \geq f)$ where $F \sim F_{m-1, n-1}$	$p = P(F \leq f)$ where $F \sim F_{m-1, n-1}$

Example 9.11: A survey was done on time for delivery of orders of pizza. Random assignment of consumers: 25 people each in group A (ordered pizza from chain A) & Group B (ordered pizza from chain B). $s_1 = 11$ minutes $s_2 = 9$ minutes. Do these data support the claim that one chain is more consistent in delivery time as than the other chain? Test using $\alpha = 0.01$.

1) $H_0: \sigma_1^2 = \sigma_2^2, H_1: \sigma_1^2 \neq \sigma_2^2$

(没特别指明是哪个连锁店的配送时间更稳定，故对立假设选择不等)

2) $\alpha = 0.01$

3) Test statistic: $F = \frac{s_1^2}{s_2^2} \sim F(m-1, n-1)$

Rejection region:

$$f \geq F_{\frac{\alpha}{2}, m-1, n-1} = F_{0.005, 24, 24} = 2.967 \text{ or } f \leq \frac{1}{F_{\frac{\alpha}{2}, n-1, m-1}} = \frac{1}{F_{0.005, 24, 24}} = 0.337$$

4) $f = \frac{11^2}{9^2} = \frac{121}{81} = 1.494$

$$0.337 < 1.494 < 2.967 \rightarrow \text{Do not reject } H_0$$

Using R, $> \text{pf}(1.494, 24, 24)$ 0.8339702

Thus, $P(F < 1.49) = 0.834, P(F > 1.49) = 0.166$

5) $p\text{-value} = 2(0.166) = 0.332$

Example 9.12: For the data in Example 9.11, find a 90% confidence interval for the ratio between the two standard deviations.

A 90% CI for σ_1^2 / σ_2^2 :

$$\left(\frac{s_1^2/s_2^2}{F_{\frac{\alpha}{2}, m-1, n-1}}, \frac{s_1^2}{s_2^2} F_{\frac{\alpha}{2}, n-1, m-1} \right) = \left(\frac{s_1^2/s_2^2}{F_{0.05, 24, 24}}, \frac{s_1^2}{s_2^2} F_{0.05, 24, 24} \right) \\ = \left(\frac{121/81}{1.984}, \frac{121}{81} (1.984) \right) = (0.753, 2.964)$$

A 90% CI for σ_1/σ_2 is

$$(\sqrt{0.753}, \sqrt{2.964}) = (0.868, 1.722)$$

Using R, suppose x and y are the observations in the two samples. Then the test can be done by

```
>var.test(x, y, conf.level=0.99)
```

Example 9.13: A class of 24 students is randomly divided into two groups and each group is taught according to a different method. Their test scores at the end of the semester show the following characteristics:

$$m = 13, \bar{x} = 74.5, s_1^2 = 82.6, n = 11, \bar{y} = 71.8, s_2^2 = 112.6$$

(a) Assume $\sigma_1^2 = \sigma_2^2$, find a 95% CI for $\mu_1 - \mu_2$.

$$s_p^2 = \frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2} = \frac{12(82.6) + 10(112.6)}{22} = 96.24$$

$$t_{\frac{\alpha}{2}, m+n-2} = t_{0.025, 22} = 2.074$$

$$\begin{aligned} \bar{x} - \bar{y} \pm t_{\frac{\alpha}{2}, m+n-2} s_p \sqrt{\frac{1}{m} + \frac{1}{n}} &= 74.5 - 71.8 \pm (2.074) \sqrt{96.24} \sqrt{\frac{1}{13} + \frac{1}{11}} \\ &= 2.7 \pm 8.335 \end{aligned}$$

A 95% confidence interval is $(-5.635, 11.035)$.

Example 9.13 (continued)

(b) Find a 98% CI for σ_1/σ_2 .

$$\begin{aligned} \left(\frac{s_1^2/s_2^2}{F_{\frac{\alpha}{2}, m-1, n-1}}, \frac{s_1^2}{s_2^2} F_{\frac{\alpha}{2}, n-1, m-1} \right) &= \left(\frac{s_1^2/s_2^2}{F_{0.01, 12, 10}}, \frac{s_1^2}{s_2^2} F_{0.01, 10, 12} \right) \\ &= \left(\frac{82.6}{(112.6)(4.71)}, \frac{82.6}{112.6} (4.30) \right) = (0.156, 3.154) \end{aligned}$$

A 98% confidence interval for σ_1/σ_2 is

$$(\sqrt{0.156}, \sqrt{3.154}) = (0.395, 1.776)$$

(c) Does the assumption of $\sigma_1^2 = \sigma_2^2$ in part (a) seem justified?

Since 1 is included in the confidence interval of (0.395, 1.776) obtained in part (b), the assumption seems to be justified.

两样本正态总体的假设检验

原假设 H_0	对立假设 H_1	检验统计量及分布	拒绝域
$\mu_1 = \mu_2$ $(\sigma_1^2 = \sigma_2^2 = \sigma^2 \text{未知})$	$\begin{cases} \mu_1 \neq \mu_2 \\ \mu_1 > \mu_2 \\ \mu_1 < \mu_2 \end{cases}$	$T = \frac{\bar{X} - \bar{Y}}{S_w \sqrt{\frac{1}{m} + \frac{1}{n}}}$ $\sim t_{m+n-2}$	$\begin{cases} T \geq t_{m+n-2}(\alpha/2) \\ T \geq t_{m+n-2}(\alpha) \\ T \leq -t_{m+n-2}(\alpha) \end{cases}$
$\sigma_1^2 = \sigma_2^2$	$\begin{cases} \sigma_1^2 \neq \sigma_2^2 \\ \sigma_1^2 > \sigma_2^2 \\ \sigma_1^2 < \sigma_2^2 \end{cases}$	$F = \frac{S_1^2}{S_2^2}$ $\sim F_{m-1, n-1}$	$\begin{cases} F \geq F_{m-1, n-1}(\alpha/2) \text{或者} \\ F \leq F_{m-1, n-1}(1 - \alpha/2) \\ F \geq F_{m-1, n-1}(\alpha) \\ F \leq \frac{1}{F_{n-1, m-1}(\alpha)} \end{cases}$